

The critique of scalar benchmark rankings as proxies for model quality is now well-established (Raji et al., 2021; Wallach et al., 2025; Bean et al., 2025; Gehrmann et al., 2021).

What the field still lacks is a shared structural vocabulary for comparing, combining, and contextualizing metric design choices. This paper provides that vocabulary: a **four-primitive typology**—**representation** (ϕ), **comparison** (D), **aggregation** (A), and **context** (C)—under which existing metrics (BLEU, BERTScore, nDCG, LLM-as-judge, calibration scores, agentic outcome measures) are explicit parameterizations of a common form.

This typology is paired with a **measurement–decision split**: metrics are noisy estimators of latent constructs, and model selection is context-dependent Pareto optimization over construct estimates, not over raw scores. The typology makes implicit metric assumptions comparable and debatable rather than hidden inside a single number.

$$M_i(x, y; \mathbf{C}) = A_i[D_i(\phi_i(x, \mathbf{C}), \phi_i(y, \mathbf{C})); \mathbf{C}]$$

ϕ

Representation

How outputs are encoded — e.g., tokens, embeddings, distributions, trajectories, etc.

D

Comparison

Distance or divergence — e.g., n-gram, cosine, entailment, learned score, etc.

A

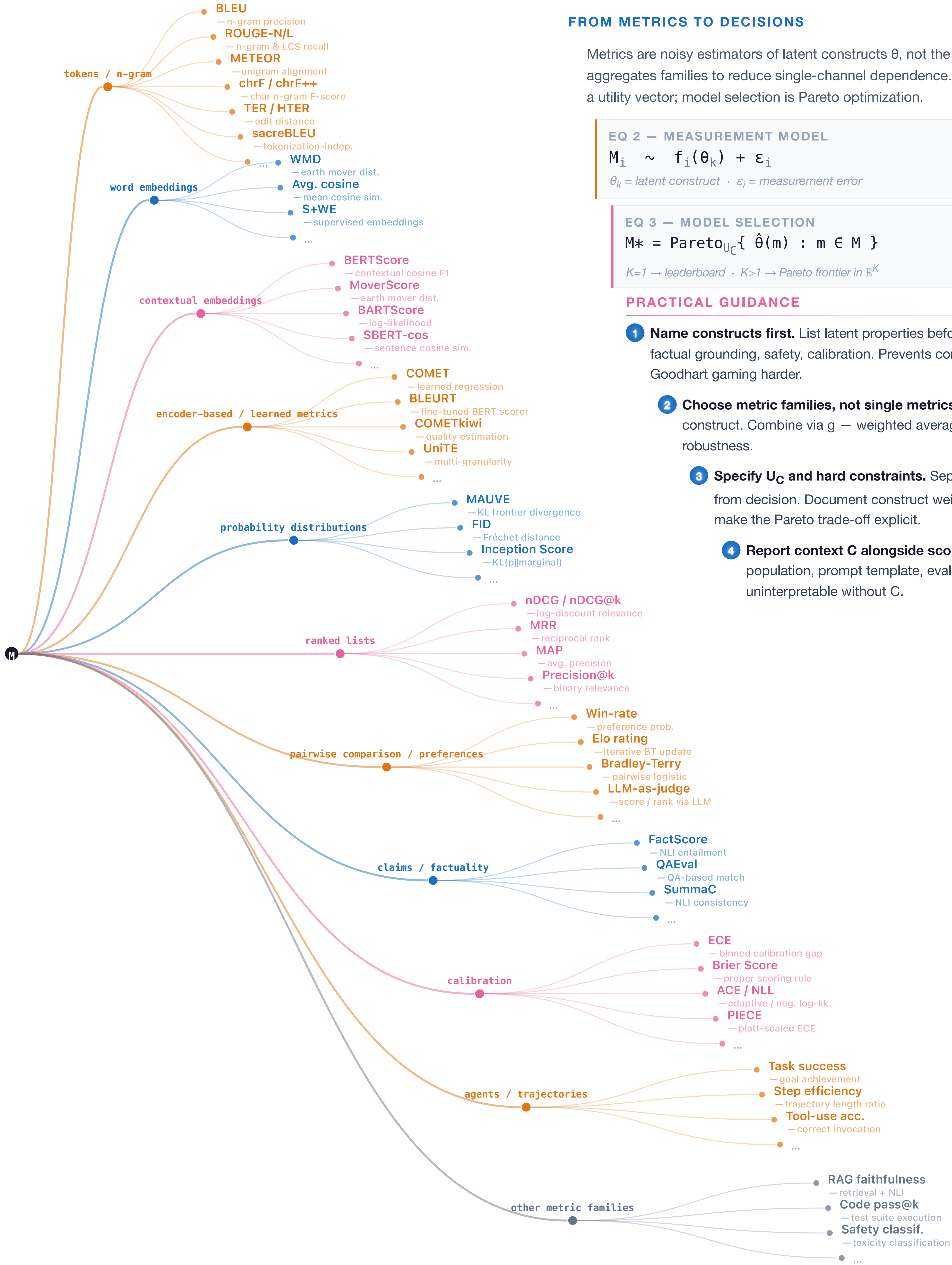
Aggregation

How signals combine — e.g., mean, F1, geometric mean, ranking, expectation, etc.

C

Context

Task, environment, annotators — e.g., conditions any stage, not a sequential step, etc.



FROM METRICS TO DECISIONS

Metrics are noisy estimators of latent constructs θ , not the objectives. A combiner g aggregates families to reduce single-channel dependence. Deployment context defines a utility vector; model selection is Pareto optimization.

EQ 2 — MEASUREMENT MODEL

$$M_i \sim f_i(\theta_k) + \epsilon_i$$

$\theta_k = \text{latent construct}$ · $\epsilon_i = \text{measurement error}$

EQ 3 — MODEL SELECTION

$$M^* = \text{Pareto}_{U_C} \{ \hat{\theta}(m) : m \in M \}$$

$K=1 \rightarrow \text{leaderboard}$ · $K>1 \rightarrow \text{Pareto frontier in } \mathbb{R}^K$

PRACTICAL GUIDANCE

- 1 Name constructs first.** List latent properties before picking metrics — fidelity, factual grounding, safety, calibration. Prevents construct mismatch; makes Goodhart gaming harder.
- 2 Choose metric families, not single metrics.** Select diverse ϕ and D per construct. Combine via g — weighted average for transparency; IRT-style for robustness.
- 3 Specify U_C and hard constraints.** Separate measurement from decision. Document construct weights and safety thresholds to make the Pareto trade-off explicit.
- 4 Report context C alongside scores.** Task distribution, annotator population, prompt template, eval model version. Results are uninterpretable without C.

Evaluation is **measurement** of latent system properties followed by **context-dependent decision**.

The (ϕ , D, A, C) typology (Eq. 1) and the **measurement–decision split** (Eqs. 2–3) expose the implicit choices inside every metric and every leaderboard. The position taken here is unambiguous: evaluation practice must state these choices openly—construct definitions, metric families, combiner functions, utility weights, and deployment context—so that trade-offs can be debated and reproduced rather than obscured in a single number.



FULL PAPER (PDF)